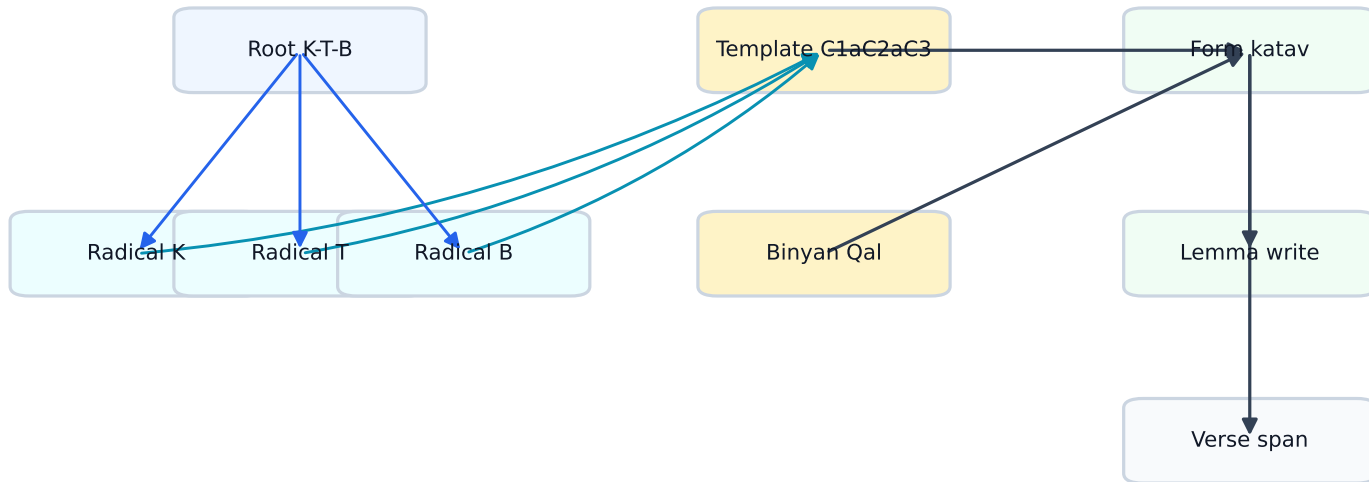


Root-template morphology as a typed graph



Records preserve analyzer uncertainty, paradigm edges, source spans, and semiring alignment losses.